

LECTURE 5: MATH1061/7861
 TOPICS: VALID AND INVALID ARGUMENTS
 METHODS FOR DETERMINING VALIDITY
 DATE: WEDNESDAY MARCH 4, 2026

Learning objectives:

- (1) Understand the difference between a valid argument and an invalid argument.
- (2) Be able to use a truth table to determine whether an argument is valid.
- (3) Use rules of inference to determine the validity of an argument.

Rules of inference

Modus Ponens	$p \rightarrow q$ p $\therefore q$	Elimination	a. $p \vee q$ $\sim q$ $\therefore p$ b. $p \vee q$ $\sim p$ $\therefore q$
<u>Modus Tollens</u>	$p \rightarrow q$ $\sim q$ $\therefore \sim p$	Transitivity	$p \rightarrow q$ $q \rightarrow r$ $\therefore p \rightarrow r$
<u>Generalization</u>	a. p $\therefore p \vee q$ b. q $\therefore p \vee q$	Proof by Division into Cases	$p \vee q$ $p \rightarrow r$ $q \rightarrow r$ $\therefore r$
Specialization	a. $p \wedge q$ $\therefore p$ b. $p \wedge q$ $\therefore q$		
Conjunction	p q $\therefore p \wedge q$	Contradiction Rule	$\sim p \rightarrow c$ $\therefore p$

Exercise: ~~Pro~~ Show that these
are all valid argument.

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Is the following argument valid?

- ① $P \rightarrow q$] Premises
② P]
 $\therefore q$] Conclusion

So the question is the following:

Assume $P \rightarrow q$ and P are both true.

Then does that mean that q is true?

Method 1: Truth Table:

P	q	$P \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

← only row where
Premises are true.

This implies q
is also true.

So the argument
is valid.

(3)
Method 2: Assume premises are true
but the conclusion is false.

If we reach a contradiction, that
implies argument is valid. Otherwise,
argument is invalid.

Suppose $p \rightarrow q$ and p are true.

But q is false.

Note: Since q is false and $p \rightarrow q$
is true, this implies p is false.

But we had assumed p is true.

So this is a contradiction.

$\Rightarrow$ Argument is valid.

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Q1:

Is the following argument valid or invalid?

$$\begin{array}{l} \cdot r \rightarrow p \wedge q \\ \cdot p \vee r \rightarrow q \\ \cdot q \rightarrow p \end{array} \left. \vphantom{\begin{array}{l} \cdot r \rightarrow p \wedge q \\ \cdot p \vee r \rightarrow q \\ \cdot q \rightarrow p \end{array}} \right\} \text{Premises}$$
$$\therefore p \leftarrow \text{Conclusion.}$$

Method 2: Suppose all premises are true but conclusion is false, i.e. p is False.

Then q is false (because $q \rightarrow p$ is True)

$p \vee r$ is also false (because $p \vee r \rightarrow q$ is True).

$p \wedge q$ is false (because p is false)

So r is false (because $r \rightarrow p \wedge q$ is True).

There is no contradiction.

So the argument is invalid.

Challenge: Use Rules of inference to
Show argument is² invalid.

Method 1: Truth Table



p	q	r	$r \rightarrow (p \wedge q)$	$(p \vee r) \rightarrow q$	$q \rightarrow p$	All Premises True?	Conclusion p
T	T	T	T	T	T	T	T
T	T	F	T	T	T	T	T
T	F	T	F	F	T	F	T
T	F	F	T	F	T	F	T
F	T	T	F	T	F	F	F
F	T	F	T	T	F	F	F
F	F	T	F	T	T	F	F
F	F	F	T	T	T	T	F

argument is invalid.

Solution to the challenge:

The rules of inference can only prove that an argument is valid. ~~you cannot~~

To prove an argument is invalid, you need to describe a valuation for which the premises are true but the conclusion is false.

(i.e. you need to use method 1 or 2 above.)